Class 11: AlphaFold

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Background

We saw last class that the main repository for biomolecular structure (PDB database) only has ~250,000 entries.

UnitProtKB (the main protein seqence database) has over 200 million entries!

Alphafold

In this hands-on session we will utilize AlphaFold to predict protein structure from sequence (Jumper et al. 2021).

Without the aid of such approaches, it can take years of expensive laboratory work to determine the structure of just one protein. With AlphaFold we can now accurately compute a typical protein structure in as little as ten minutes.

The EBI Alphafold database

The EBI Alphafold database contains lots of computed structure models. It is increasingly likely that the structure you are interested in is already in this database at https://alphafold.ebi.ac.uk/

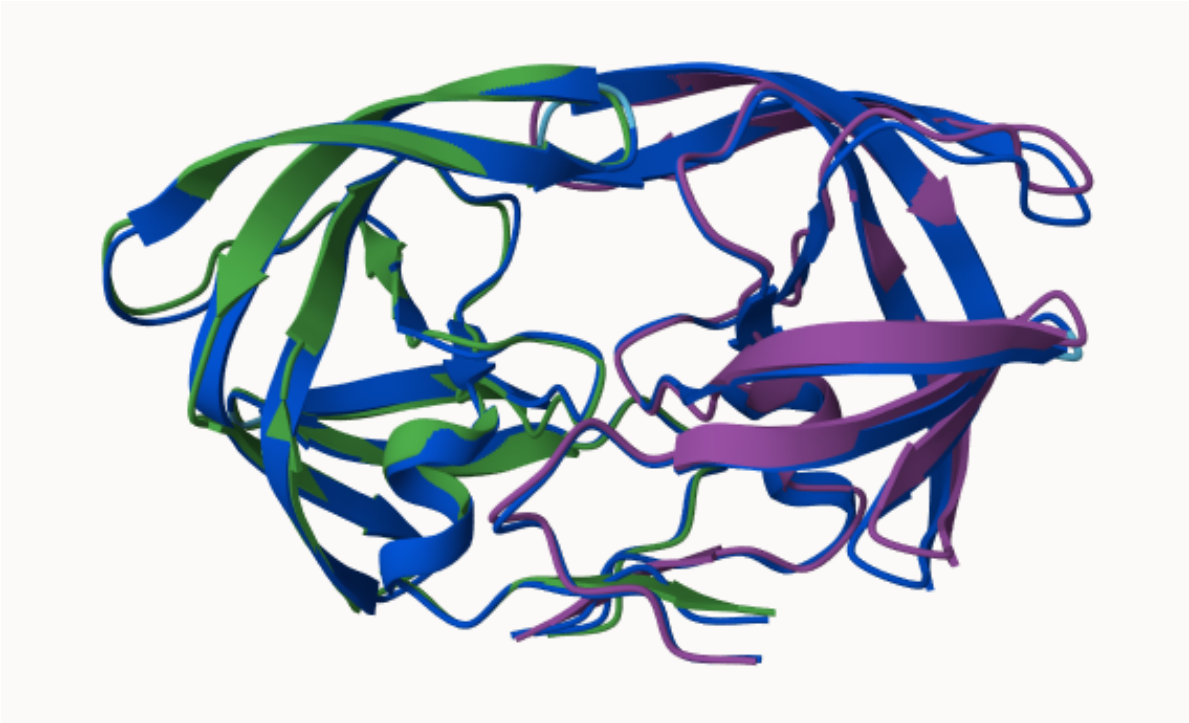
There are 3 major outputs from AlphaFold

1. A model of structure in **PDB** format
2. A **pLDDT score**: that tells us how confident the model is for a given residue in your proein (High values are better, above 70)
3. a **PAE score** that tells us about protein packing quality

If you can't find a matching entry for the sequence you are interested in AFDB you can run AlphaFold yourself…

Running AlphaFold

We will use ColabFold to run AlphaFold on our sequence < https://github.com/sokrypton/ColabFold



##Interpreting Results

We can read all the AlphaFold results into R and do more quantitative analysis than just viewing the structures in Molstar:

Read all the PDB models:

```
library(bio3d)
p <- read.pdb("hivpr_23119.result/hivpr_23119/hivpr_23119_unrelaxed_rank_001_alphafold2_multi
```

```
pdb_files <- list.files("hivpr_23119.result/hivpr_23119/", pattern = ".pdb", full.names = T)
pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

```
Reading PDB files:
hivpr_23119.result/hivpr_23119/hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_2
hivpr_23119.result/hivpr_23119/hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_4
hivpr_23119.result/hivpr_23119/hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_1
hivpr_23119.result/hivpr_23119/hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_5
```

```
hivpr_23119.result/hivpr_23119/hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_
.....

Extracting sequences

pdb/seq: 1   name: hivpr_23119.result/hivpr_23119/hivpr_23119_unrelaxed_rank_001_alphafold2_m
pdb/seq: 2   name: hivpr_23119.result/hivpr_23119/hivpr_23119_unrelaxed_rank_002_alphafold2_m
pdb/seq: 3   name: hivpr_23119.result/hivpr_23119/hivpr_23119_unrelaxed_rank_003_alphafold2_m
pdb/seq: 4   name: hivpr_23119.result/hivpr_23119/hivpr_23119_unrelaxed_rank_004_alphafold2_m
pdb/seq: 5   name: hivpr_23119.result/hivpr_23119/hivpr_23119_unrelaxed_rank_005_alphafold2_m
```

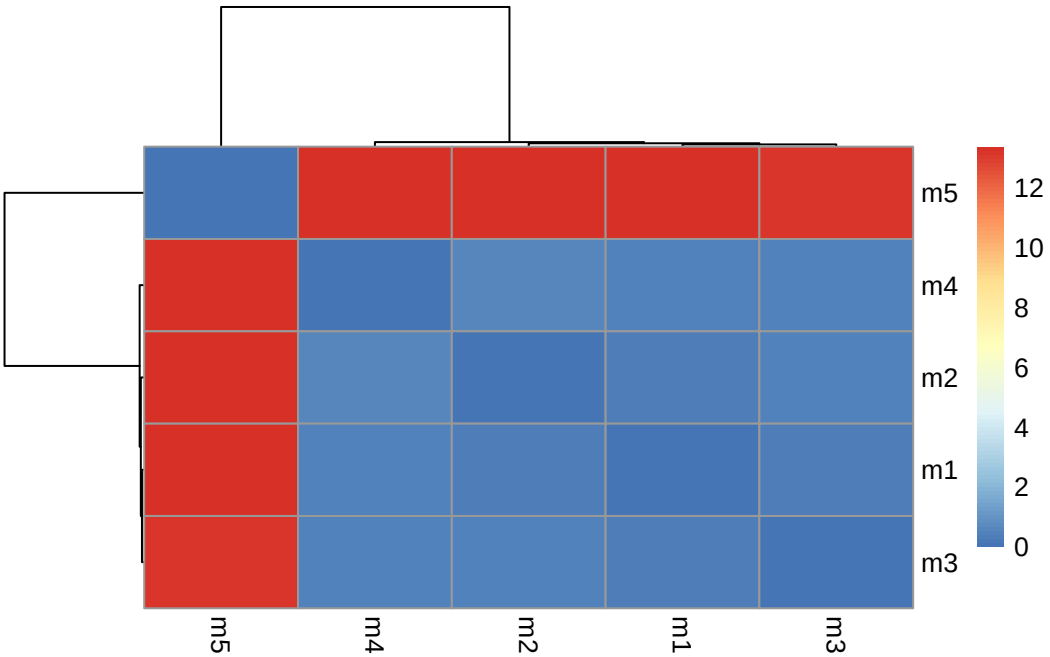
```r
#library(bio3dview)
#view.pdbs(pdbs)
```

How similar or different are my models?

```r
rd <- rmsd(pdbs)
```

```
Warning in rmsd(pdbs): No indices provided, using the 198 non NA positions
```

```r
library(pheatmap)
colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```



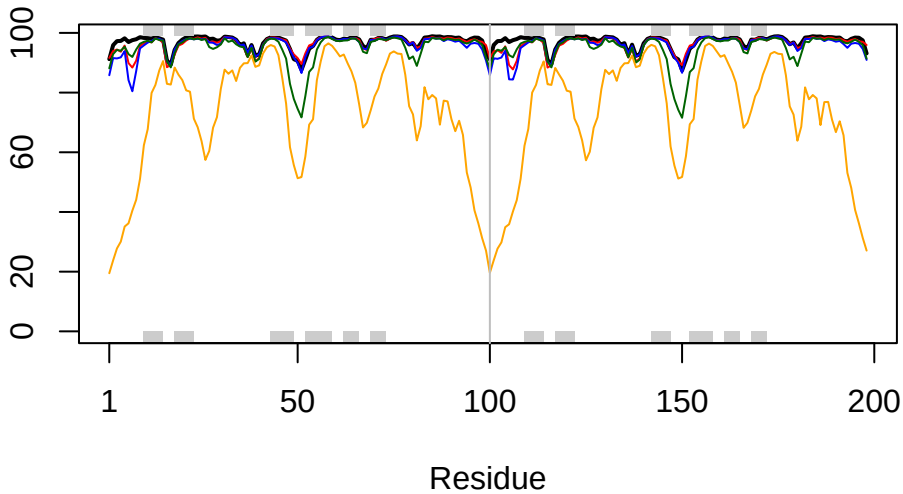
Left off here from class.

Plotting pLDDT values across the modles:

```
pdb <- read.pdb("1hsg")
```

```
  Note: Accessing on-line PDB file
```

```
plotb3(pdbs$b[1,], typ="l", lwd=2, sse=pdb)
points(pdbs$b[2,], typ="l", col="red")
points(pdbs$b[3,], typ="l", col="blue")
points(pdbs$b[4,], typ="l", col="darkgreen")
points(pdbs$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```



Using core.find() function to find the most consistent rigid core which is common between the models, which can be used to improve the fitting of the models.

```
core <- core.find(pdbs)
```

```
core size 197 of 198  vol = 58.625
core size 196 of 198  vol = 52.96
core size 195 of 198  vol = 49.614
core size 194 of 198  vol = 46.269
core size 193 of 198  vol = 44.802
core size 192 of 198  vol = 43.508
core size 191 of 198  vol = 42.246
core size 190 of 198  vol = 40.982
core size 189 of 198  vol = 39.788
core size 188 of 198  vol = 38.576
core size 187 of 198  vol = 37.347
core size 186 of 198  vol = 36.342
core size 185 of 198  vol = 35.226
core size 184 of 198  vol = 34.286
core size 183 of 198  vol = 33.532
core size 182 of 198  vol = 32.75
core size 181 of 198  vol = 32.195
core size 180 of 198  vol = 31.494
```

```
core size 179 of 198  vol = 31.013
core size 178 of 198  vol = 30.448
core size 177 of 198  vol = 29.837
core size 176 of 198  vol = 29.536
core size 175 of 198  vol = 29.483
core size 174 of 198  vol = 29.556
core size 173 of 198  vol = 29.367
core size 172 of 198  vol = 29.366
core size 171 of 198  vol = 29.271
core size 170 of 198  vol = 29.272
core size 169 of 198  vol = 29.032
core size 168 of 198  vol = 28.853
core size 167 of 198  vol = 28.495
core size 166 of 198  vol = 28.045
core size 165 of 198  vol = 27.445
core size 164 of 198  vol = 26.7
core size 163 of 198  vol = 26.022
core size 162 of 198  vol = 25.412
core size 161 of 198  vol = 24.67
core size 160 of 198  vol = 24.117
core size 159 of 198  vol = 23.679
core size 158 of 198  vol = 23.089
core size 157 of 198  vol = 22.457
core size 156 of 198  vol = 22.038
core size 155 of 198  vol = 21.617
core size 154 of 198  vol = 21.215
core size 153 of 198  vol = 20.757
core size 152 of 198  vol = 19.861
core size 151 of 198  vol = 19.611
core size 150 of 198  vol = 19.177
core size 149 of 198  vol = 18.875
core size 148 of 198  vol = 18.447
core size 147 of 198  vol = 17.829
core size 146 of 198  vol = 17.441
core size 145 of 198  vol = 17.063
core size 144 of 198  vol = 16.662
core size 143 of 198  vol = 16.4
core size 142 of 198  vol = 15.791
core size 141 of 198  vol = 15.113
core size 140 of 198  vol = 14.581
core size 139 of 198  vol = 14.037
core size 138 of 198  vol = 13.63
core size 137 of 198  vol = 12.821
```

```
core size 136 of 198  vol = 12.496
core size 135 of 198  vol = 12.232
core size 134 of 198  vol = 11.95
core size 133 of 198  vol = 11.705
core size 132 of 198  vol = 11.36
core size 131 of 198  vol = 10.983
core size 130 of 198  vol = 10.545
core size 129 of 198  vol = 9.983
core size 128 of 198  vol = 9.562
core size 127 of 198  vol = 9.27
core size 126 of 198  vol = 8.703
core size 125 of 198  vol = 8.091
core size 124 of 198  vol = 7.644
core size 123 of 198  vol = 7.291
core size 122 of 198  vol = 6.936
core size 121 of 198  vol = 6.575
core size 120 of 198  vol = 6.114
core size 119 of 198  vol = 5.689
core size 118 of 198  vol = 5.281
core size 117 of 198  vol = 4.892
core size 116 of 198  vol = 4.582
core size 115 of 198  vol = 4.307
core size 114 of 198  vol = 4.056
core size 113 of 198  vol = 3.72
core size 112 of 198  vol = 3.561
core size 111 of 198  vol = 3.409
core size 110 of 198  vol = 3.276
core size 109 of 198  vol = 3.035
core size 108 of 198  vol = 2.97
core size 107 of 198  vol = 2.859
core size 106 of 198  vol = 2.784
core size 105 of 198  vol = 2.484
core size 104 of 198  vol = 2.279
core size 103 of 198  vol = 2.203
core size 102 of 198  vol = 2.111
core size 101 of 198  vol = 2.002
core size 100 of 198  vol = 1.905
core size 99 of 198  vol = 1.788
core size 98 of 198  vol = 1.673
core size 97 of 198  vol = 1.526
core size 96 of 198  vol = 1.403
core size 95 of 198  vol = 1.165
core size 94 of 198  vol = 0.987
```

```
 core size 93 of 198  vol = 0.803
 core size 92 of 198  vol = 0.727
 core size 91 of 198  vol = 0.587
 core size 90 of 198  vol = 0.525
 core size 89 of 198  vol = 0.396
 FINISHED: Min vol ( 0.5 ) reached
```

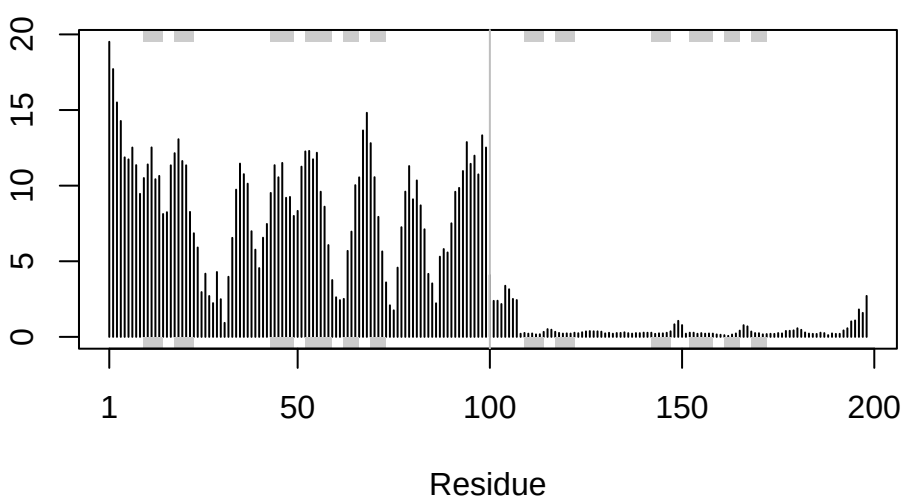
```r
core.inds <- print(core, vol=0.5)
```

```
# 90 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     9  49     41
2    52  78     27
3    80  97     18
```

```r
xyz <- pdbfit(pdbs, core.inds, outpath="corefit_structures")
```

```r
rf <- rmsf(xyz)

plotb3(rf, sse=pdb)
abline(v=100, col="gray", ylab="RMSF")
```



From the model, the first chain varies a lot between models while the second chain varies.

Predicted Alignment Error for domains

```
results_dir <- "hivpr_23119/"
library(jsonlite)
pae_files <- list.files(path=results_dir,
                        pattern=".*model.*\\.json",
                        full.names = TRUE) ## reads and lists json files within the results
```

```
pae1 <- read_json(pae_files[1],simplifyVector = TRUE)
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)
attributes(pae1)
```

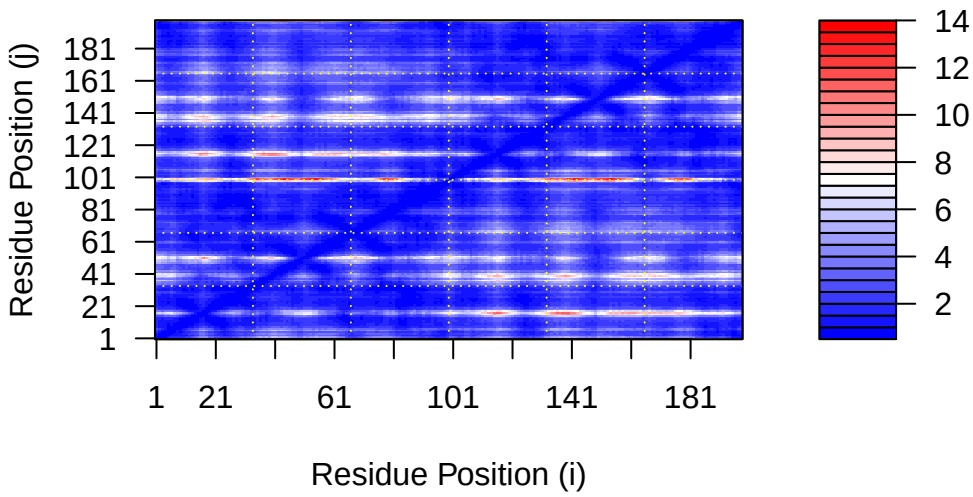
```
$names
[1] "plddt"   "max_pae" "pae"     "ptm"     "iptm"
```

```
head(pae1$plddt) ##gives per residue plddt scores
```

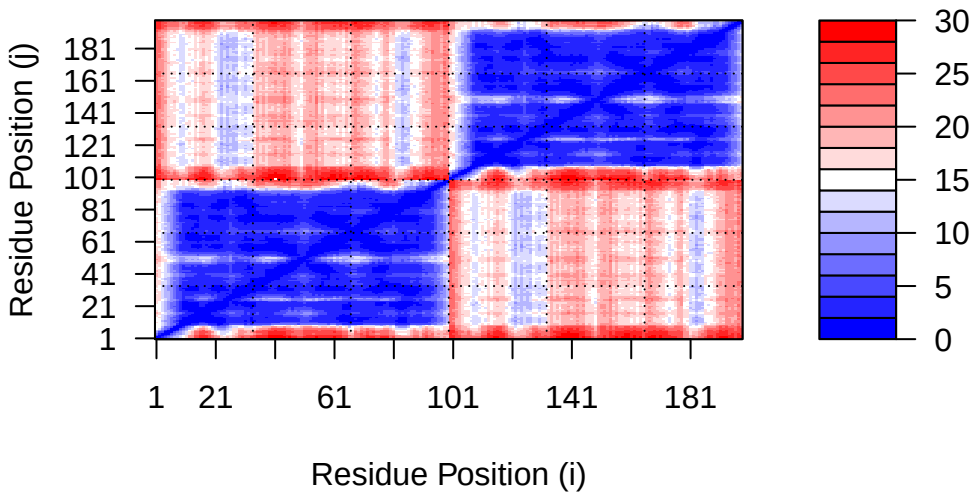
```
[1] 91.19 95.88 97.25 97.31 98.19 96.94
```

Plotting residue by residue PAE scores:

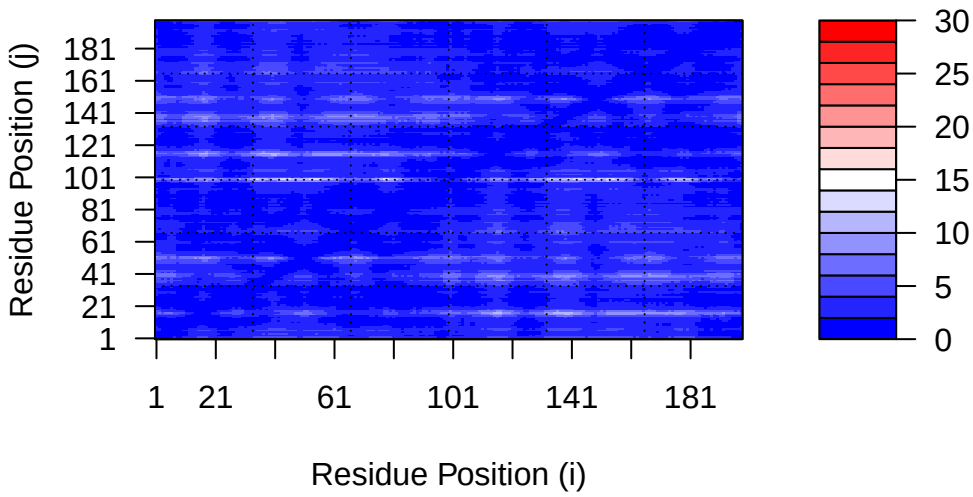
```
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)")
```




```r
plot.dmat(pae5$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30)) ##plot for model 5
```




```
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30)) ##plot for model 1
```



Residue conservation from alignment file

```
aln_file <- list.files(path=results_dir,
                       pattern=".a3m$",
                        full.names = TRUE)
aln_file
```

```
[1] "hivpr_23119/hivpr_23119.a3m"
```

```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

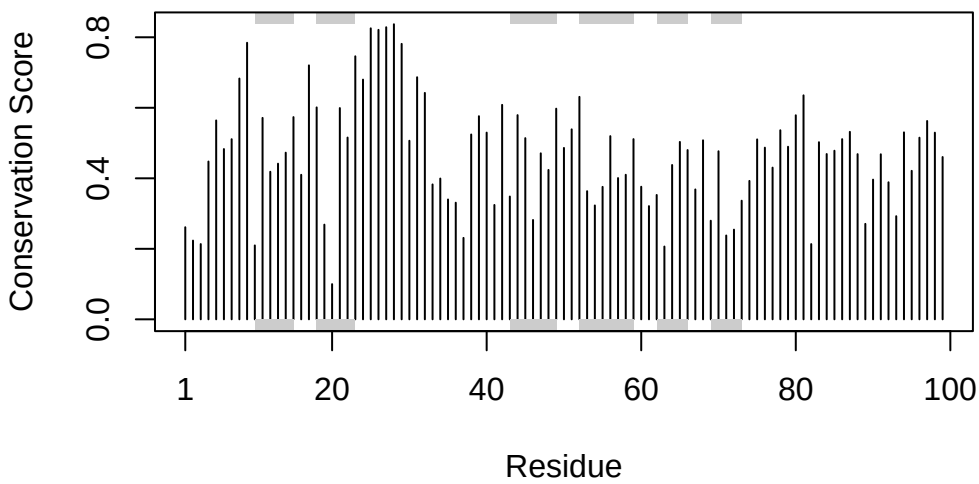
```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

```
dim(aln$ali) ##returns number of sequences in the alignment
```

```
[1] 5397  132
```

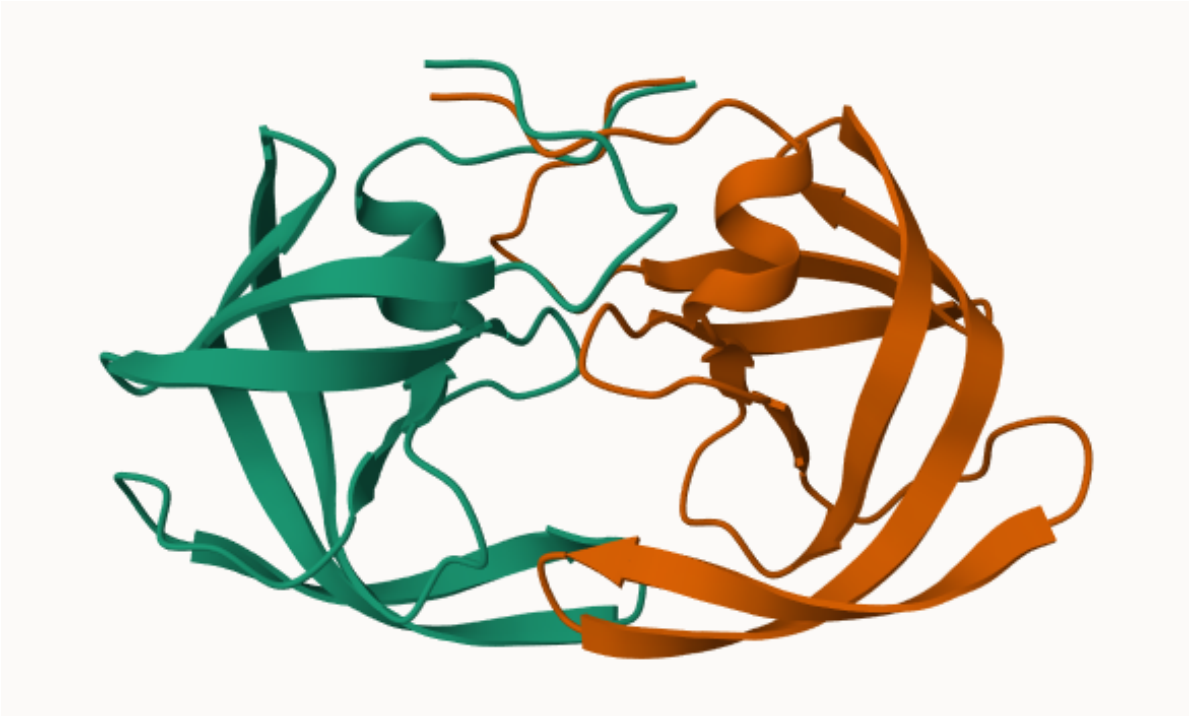
Score residue conservation in the alignment using conserv():

```
sim <- conserv(aln)
plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```



Residues 25-28 have the highest conservation score, we can generate a consensus sequence with a high cutoff value to eliminate other residues with a smaller score:

```
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
  [1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
 [37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

```r
m1.pdb <- read.pdb(pdb_files[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```